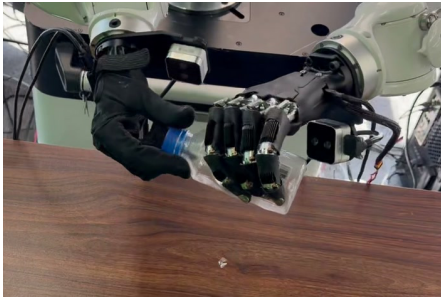


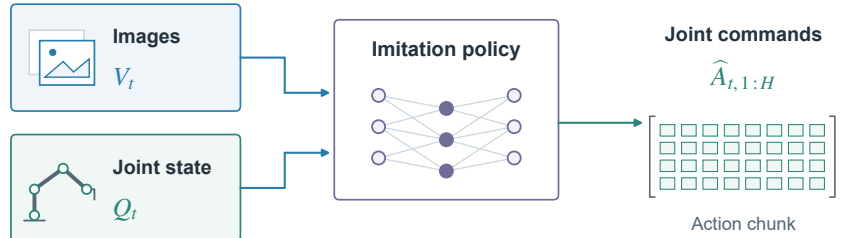
Blind Hands: Visual Reliance and Feedback Error in Dexterous Imitation

Anonymous submission

(a) Bottle-cap manipulation



(b) Which inputs guide the action?



Task success alone does not identify input use.

Fig. 1. Task and research question. (a) The robot grasps the attached cap before turning it. (b) Images and joint states are available to the imitation policy, but task completion alone does not identify how they influence commands. We diagnose input dependence and evaluate the effects of changing it.

Abstract—What information does an imitation-learning policy actually use to generate actions? Task success alone does not establish whether the policy uses visual scene information or exploits predictable joint-state correlations. We investigate this question in bimanual dexterous manipulation through visual and joint-state interventions, using tactile contact events to anchor exploratory diagnostics. We find weak image dependence in the diagnosed policy, including continued progression under frozen images in recorded-stream replay. Architectural and contact-aligned audits rule out explicit action-history copying and absent joint-state variation at contact onset. To address this visual underuse, we study action reparameterization, state noise, and dropout, asking whether encouraging visual dependence also improves command accuracy and feedback behavior. Experimental results from a controlled three-seed study show that, under full-state evaluation, 50% state dropout raises visual reliance from 0.392 to 0.990 relative to no dropout, while increasing command RMSE by 8.4% and left-hand numerical replay amplification from 2.09 to 15.97. By separating encoder state from command-composition state, our analysis identifies a feedback path retained when encoder state is absent and shows that exact relative targets preserve visual information conditional on state. Complementary noise and transformer comparisons reveal different response-error patterns. Together, these findings show why identifying input dependence and evaluating the consequences of changing it require joint measurements of visual response, command fidelity, and feedback error.

I. INTRODUCTION

What information does an imitation policy use? Learning from demonstrations [1], [2] has progressed from image-to-control systems [3] to learned perception and control [4], [5]. Images describe objects and their relation to the hands; proprioception describes robot configuration. Task completion does not identify how these inputs produced the outcome. Dense demonstrations also contain strong state–command correlations, raising the question of how to distinguish visual

response from dependence on predictable movement regularities.

Causal confusion [6], action copying [7], and uneven learning across task phases [8] expose related risks. Recent studies report proprioceptive overdependence and poor spatial generalization [9], or limited visual contribution during motion transitions [10]. These findings motivate diagnosing input use before selecting an intervention. We call the weak visual response examined here *blind hands*.

We study policies based on implicit maximum likelihood estimation (IMLE) [11] and reproduce Action Chunking with Transformers (ACT) [12]. Visual/state interventions and tactile contact events anchor our initial diagnosis of an IMLE-based bottle-manipulation policy. The initial policy responds weakly to image changes and advances with frozen images in recorded-stream replay. Architecture and data audits exclude explicit action-history copying and absent contact-onset variation; state perturbations constrain strong servo and cross-limb-clock explanations. This diagnosis motivates attempts to strengthen visual guidance.

Changing input dependence also affects command generation and feedback: state conditions prediction, supplies the reference added to relative actions, and can support correction when earlier errors re-enter the policy. Within the IMLE-based family, a controlled three-seed dropout study finds increased visual reliance alongside higher command error and left-limb amplification. Complementary consistency-noise comparisons and our ACT reproduction, testing absolute/relative targets at two chunk lengths, reveal different response–error patterns. Our analysis separates the retained composition path from learned state dependence.

We contribute a targeted diagnosis of input dependence; a controlled, three-seed counterexample to selecting state

interventions by visual reliance alone; and an analysis relating input dependence, command composition, and feedback error. These connect what a policy responds to with the consequences of changing that response.

II. RELATED WORK

Imitation and dexterous manipulation. Generative imitation includes energy-based policies [13], discrete action modes [14], diffusion [15], and IMLE generators [11]. Dexterous learning uses demonstrations with reinforcement learning [16], human-video transfer [17], and interfaces such as Holo-Dex and AnyTeleop [18], [19]. ALOHA uses ACT; Mobile ALOHA and UMI extend bimanual learning and policy interfaces [12], [20], [21]. Our question concerns the information used by a learned policy and the effects of intervening on that use.

Input dependence and shortcuts. Shared multimodal representations [22] can have unequal optimization dynamics [23], [24]. Shortcut learning [25], causal confusion [6], and action copying [7] explain reliance on predictive correlates; keyframe learning emphasizes consequential transitions [8]. State-free Policy removes proprioception with relative end-effector actions and sufficient task-relevant visual coverage [9]. GAP modulates proprioceptive gradients during motion transitions [10]. We examine joint-space dexterous policies, including the command-composition path that remains after withholding encoder state.

Data and evaluation. RoboTurk, BridgeData V2, and DROID support demonstration collection [26]–[28]; robosuite and RLBench provide controlled environments [29], [30]. Dataset quality and algorithm choice affect offline learning [31]. RT-1, Octo, and OpenVLA explore learning across diverse robot data [32]–[34]. We report within-collection comparisons and distinguish episode uncertainty from training-run variation [35], [36].

Perturbations and feedback. Dropout and input noise regularize predictors [37], [38]. DART adds noise during demonstration collection to elicit correction [39]; our noise case perturbs recorded state and recomputes relative targets. Distribution shift and compounding error are central to imitation [40]–[42]. Recorded-stream replay measures propagation through the learned command map under fixed external observations.

III. DIAGNOSING WHAT THE POLICY USES

A. Input interventions and contact timing

The exploratory collection contains 199 bottle-cap demonstrations: 137,473 frames at 30 Hz with joint states/actions, head RGB, and right-glove pressure. The diagnosed generator uses one observation without action history or recurrent state and predicts 16 commands. Visual/state substitutions hold other inputs fixed. Frozen-image replay tests whether progression requires visual updates.

Tactile sensing provides an event-time reference independent of the action residual. Contact-aligned audits find substantial joint-state variation at onset, excluding the strict

TABLE I

EXPLORATORY DIAGNOSTIC CHECKS ON THE INITIAL POLICY AND COLLECTION. THESE CHECKS CONSTRAIN SPECIFIC EXPLANATIONS; THEY ARE NOT INDEPENDENT TRAINING REPLICATIONS. REPORTED INTERVALS ARE FROM THE ARCHIVED PROBES.

Question	Check and finding
Explicit action copying?	One observation; no previous-action input or recurrent policy state.
No contact-onset variation?	Across-demonstration arm-state standard deviation: $16.2^\circ/\text{joint}$; 16.0° after time regression.
Unit-gain state servo?	Arm perturbation gain 0.222 [0.21, 0.24], well below the unit-gain prediction.
Strong cross-limb clock?	Arm-only effect on the hand: -0.04 [-0.19 , $+0.16$]; the strong prediction is unsupported.
Visual updates needed for progression?	Frozen-image replay continues to advance along generated movements.

no-variation explanation (Table I). Per-dimension action-changepoint enrichment differs between the right and left hands, 2.42 and 0.61, motivating separate reporting. Pressure is used here to locate contact, not to estimate the benefit of adding tactile input. Appendix III specifies sensor quality and the scope of these historical diagnostics.

B. From diagnosis to intervention

Weak image response, frozen-image progression, and weak state re-anchoring are compatible with learned movement regularities and limited state-dependent correction. They do not uniquely identify an internal representation or attribute percentages of task success to inputs.

This diagnosis motivates interventions intended to increase visual use. Historical absolute/relative comparisons reported larger image responses with relative targets, but their original baseline checkpoint and full training match are incompletely recovered. We therefore use the controlled dropout study to quantify a response–error trade-off and the recovered noise and ACT pairs to examine other settings. The exploratory findings motivate these comparisons; they do not inherit the held-out status or training replication of the later dropout experiment.

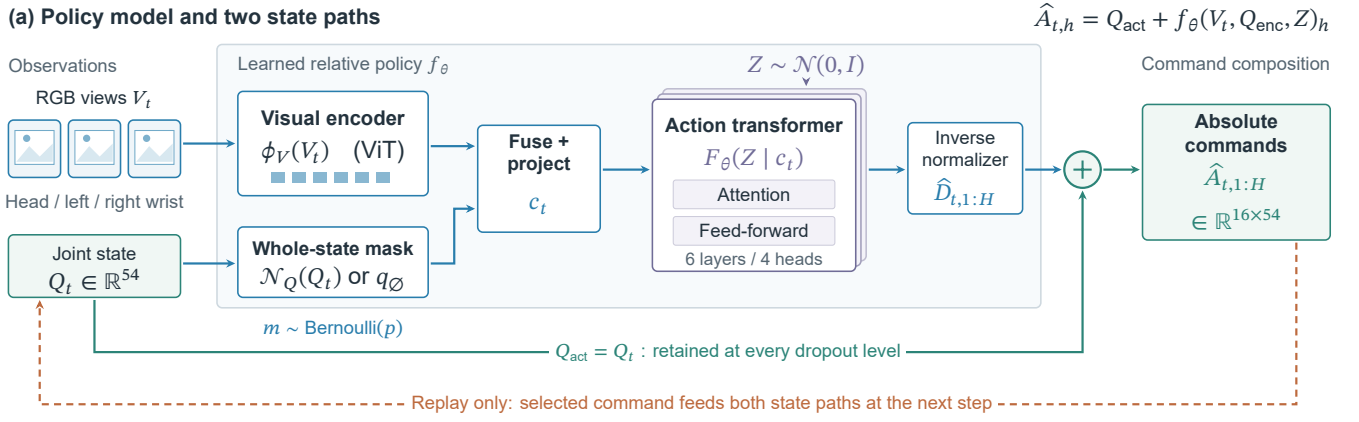
IV. INTERVENTIONS, MEASUREMENTS, AND FEEDBACK

A. Conditional action model and two state paths

The robot has two 7-DoF arms and two 20-DoF hands. In the controlled study, the policy receives three RGB views V_t and joint state $Q_t \in \mathbb{R}^{54}$, and predicts a chunk of $H = 16$ commands. Each relative target is $D_{t,h} = A_{t,h} - Q_t$: every step subtracts the same current observation state. Figure 2 depicts this model and its evaluation paths. The earlier diagnostic and noise models use their recorded sensory configurations.

A hybrid vision transformer [43] encodes images. Its features and normalized state form the condition for an action transformer [44] with Gaussian latent Z . Inverse

(a) Policy model and two state paths



(b) Joint evaluation of the composed command map g

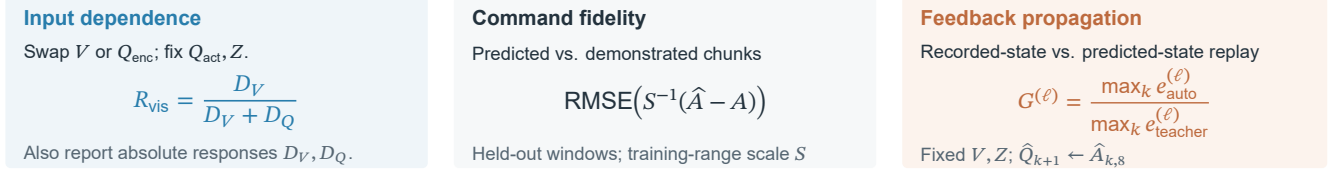


Fig. 2. Architecture and evaluation of the controlled policy. (a) Three RGB views and masked encoder state condition a stochastic action transformer. The denormalized relative chunk is added to the retained composition state Q_{act} . The dashed loop feeds a selected command into both state paths during numerical replay. (b) Input interventions, demonstration error, and paired replay assess dependence, fidelity, and propagation, respectively. Paired replay uses the same recorded images and latent draws; it does not simulate environment responses.

normalization yields relative predictions $f_\theta(V, Q_{enc}, Z)_h$ and composed commands

$$g_h(V, Q_{enc}; Q_{act}, Z) = Q_{act} + f_\theta(V, Q_{enc}, Z)_h. \quad (1)$$

Encoder state conditions prediction; composition state converts the relative output to an absolute command. They coincide in ordinary inference. Fixing composition state during intervention isolates the learned response.

B. Whole-state dropout as a controlled intervention

For each training example, a Bernoulli mask $m \sim \text{Bernoulli}(p)$ replaces the entire normalized state vector with a learned token $q_\emptyset$:

$$q_{enc} = (1 - m)\mathcal{N}_Q(Q_t) + mq_\emptyset. \quad (2)$$

Training uses $p \in \{0, 0.5, 1\}$ and a common IMLE-style objective [11], [45], [46]. Targets and composition state are unchanged, including at $p = 1$. Appendix I gives implementation details.

C. Three complementary measurements

Input response. For paired inputs, we substitute recorded images V^π or encoder state Q^π , retaining the recipient's Q_{act} and latent draw. Let S be the diagonal matrix of training command ranges and $\bar{g}_h = S^{-1}g_h$. Within each model, expectations average over windows, chunk steps h , and paired draws:

$$\begin{aligned} D_V^2 &= \mathbb{E} \|\bar{g}_h(V, Q; Q, Z) - \bar{g}_h(V^\pi, Q; Q, Z)\|^2 / 54, \\ D_Q^2 &= \mathbb{E} \|\bar{g}_h(V, Q; Q, Z) - \bar{g}_h(V, Q^\pi; Q, Z)\|^2 / 54, \\ R_{vis} &= D_V / (D_V + D_Q). \end{aligned} \quad (3)$$

This finite intervention measures command sensitivity, not task-success attribution or visual understanding. Reporting both D_V and D_Q distinguishes increased image response from decreased state response.

Command fidelity. Range-scaled root-mean-square error (RMSE) compares predicted absolute chunks with demonstrated commands on the same windows.

Feedback error. Two replay branches share recorded images and paired generator draws. The teacher-state branch uses recorded joint state; the autoregressive branch substitutes its preceding predicted command into both state paths. For each episode and limb ℓ , we compare peak trajectory errors:

$$G^{(\ell)} = \frac{\max_k e_{\text{auto}}^{(\ell)}(k)}{\max_k e_{\text{teacher}}^{(\ell)}(k)}. \quad (4)$$

Replay fixes the recorded sensory stream. Each model has its own denominator; absolute errors accompany G . The archived wrapped residual mixes recorded units (Appendix II), making G a numerical score, not a physical-angle metric. Replay excludes environment, servo, and contact responses.

D. What target subtraction changes

Exact relative targets preserve conditional visual information: $I(A - Q; V | Q) = I(A; V | Q)$. Matched functions $f_D = f_A - Q$ also give identical commands and derivatives. This coordinate change alone does not determine learned feedback or accelerate visual learning (Appendix IV).

The retained feedback path. For a fixed image, latent draw, and chunk step h , the composed map with both state

paths equal to Q has derivative

$$\frac{\partial g_h}{\partial Q} = I + \frac{\partial f_{\theta,h}}{\partial Q}. \quad (5)$$

A missing encoder token sets the learned state derivative to zero while retaining the identity term. More concretely, let Q_k^* denote recorded state and u_k the selected relative prediction under missing-state input. Under matched sensory inputs and latent draws, u_k is the same in both replay branches. With teacher residual $\eta_k = Q_k^* + u_k - Q_{k+1}^*$ and autoregressive residual $e_k = Q_k - Q_k^*$, initialization at $e_0 = 0$ gives

$$e_{k+1} = e_k + \eta_k, \quad e_k = \sum_{j < k} \eta_j. \quad (6)$$

Persistent residuals accumulate additively in ordinary coordinates, before the replay transform. Full-state models can supply learned correction. Donor sensitivity and the feedback derivative differ, motivating direct replay measurement; Appendix IV gives the bound.

V. CONSEQUENCES OF CHANGING INPUT DEPENDENCE

A. Design and evaluation scope

The controlled bottle-manipulation study uses a separate 199-episode collection from six recording stores, with head and bilateral wrist RGB and joint state/action data, and no tactile input. A common split contains 157 training and 42 held-out episodes. Each dropout level uses three training seeds. We evaluate two designated checkpoints with full encoder state (F) and the missing token (M). Supported contrasts keep availability fixed: partial versus no dropout under F, and complete versus partial dropout under M. Evaluating $p = 0$ under M or $p = 1$ under F uses an untrained branch and is treated only as a stress test.

Probes use three command-change windows per held-out episode and real donors from another held-out episode in the same store. Replay selects command eight for 25 replans. Normalizers use training data. Checkpoints, donors, and criteria were fixed before this split-trained experiment, informed by the exploratory work (Appendix II).

We report 95% paired episode-bootstrap intervals conditional on the trained models and fixed donors, and individual seed results. Post hoc leave-one-store-out analysis tests within-collection sensitivity, not new-session uncertainty.

B. Partial dropout: more visual response, higher errors

Partial dropout increases both visual reliance and absolute image response under full-state evaluation (Tables III and V). At the final checkpoint, R_{vis} rises from 0.392 to 0.990. Image response D_V increases from 0.05963 to 0.10426, while state response D_Q decreases from 0.09240 to 0.00110. The reliance difference is 0.597 with interval [0.571, 0.623].

RMSE rises from 0.05654 to 0.06127: +8.4%, ratio interval [1.054, 1.111]. The earlier ratio is 1.088 [1.057, 1.124]. Point estimates satisfy the predeclared 10% guardrail, but interval bounds exceed it, leaving noninferiority unestablished.

TABLE II

FULL-STATE $p = 0.5$ VERSUS $p = 0$, SEPARATELY BY TRAINING SEED. RELIANCE IS A DIFFERENCE; ALL OTHER ENTRIES ARE CANDIDATE/BASELINE RATIOS. LA/RA/LH/RH DENOTE LEFT/RIGHT ARM/HAND. CHECKPOINTS REUSE THE SAME TRAINING RUNS.

Epoch	Seed	ΔR	RMSE	LA	RA	LH	RH
99	42	0.631	1.121	1.85	1.26	8.21	1.06
99	43	0.630	1.103	1.80	1.04	8.14	0.68
99	44	0.453	1.043	1.34	0.93	4.99	1.24
198	42	0.633	1.099	2.13	1.21	9.26	1.00
198	43	0.632	1.093	1.97	1.09	8.49	0.97
198	44	0.527	1.059	1.43	0.93	5.98	1.17

Left-arm and left-hand amplification rise from 2.25 to 4.04 and from 2.09 to 15.97, respectively. The corresponding ratios are 1.797 [1.645, 1.977] and 7.636 [4.972, 8.286]. Right-arm and right-hand ratio intervals include one. Numerical replay flag fractions ($G \geq 3$) rise from 11.1% to 87.3% for the left arm and from 11.1% to 94.4% for the left hand. This left-limb amplification cost makes partial dropout fail the joint selection rule (Table III).

Absolute errors corroborate the amplification increase. Left-hand teacher-state peaks change from 0.300 to 0.305, while autoregressive peaks rise from 0.557 to 5.280. Left-arm teacher/auto summaries change from 8.672/18.617 to 8.786/34.679. Peaks and episode-wise ratios have separate aggregation paths (Appendix II).

C. Repeatability and complete dropout

Reliance, command error, and left-limb amplification rise in every seed at both checkpoints; right-limb changes are mixed (Table II). Checkpoints reuse the same trained models. Principal directions persist in all six store omissions. One early omission exceeds the 10% RMSE guardrail; the joint decision is unchanged.

Under missing-state evaluation, complete/partial-dropout RMSE ratios are 0.999 [0.978, 1.024] and 0.993 [0.980, 1.008]. Both intervals admit higher or lower error; two of three final-checkpoint seeds have higher error under complete dropout. Left-hand amplification remains high. Accuracy ordering and equivalence are unestablished.

D. Complementary noise and architecture comparisons

Noise and ACT pairs use the earlier collection and their own protocols. Their unbounded image/state response ratio ρ_{vis} differs from R_{vis} ; comparisons remain within each setting (Appendix III).

Consistency noise can yield a different trade-off. In the existing one-seed comparison on two training episodes, consistency noise reduces both amplification and autoregressive peak error in all eight limb-episode combinations (Table IV). The image/state response ratio rises from 0.576 to 1.534. Teacher-state errors vary across limbs, including a right-hand increase from 0.059 to 0.152 on one episode.

The pair combines state noise $\tilde{Q} = Q + \xi$ with recomputed targets $\tilde{D}_h = A_h - \tilde{Q}$, preserving $\tilde{Q} + \tilde{D}_h = A_h$. Individual effects and broader generalization are unmeasured.

TABLE III

EFFECT OF 50% VERSUS NO STATE DROPOUT UNDER FULL-STATE EVALUATION. CONTRASTS SUMMARIZE THREE TRAINING SEEDS; BOTH CHECKPOINTS REUSE THE SAME MODELS. INTERVALS ARE CONDITIONAL 95% PAIRED EPISODE-BOOTSTRAP INTERVALS.

Measure	Unit	Epoch 99		Epoch 198	
		Change	95% interval	Change	95% interval
Visual reliance (ΔR_{vis})	pp	+57.1	[+54.7, +59.2]	+59.7	[+57.1, +62.3]
Command RMSE	%	+8.8	[+5.7, +12.4]	+8.4	[+5.4, +11.1]
Replay amplification: left arm	%	+63.1	[+50.4, +76.8]	+79.7	[+64.5, +97.7]
Replay amplification: right arm	%	+7.1	[−1.3, +18.8]	+7.0	[−3.7, +16.7]
Replay amplification: left hand	%	+572.1	[+502.4, +619.3]	+663.6	[+397.2, +728.6]
Replay amplification: right hand	%	−5.0	[−9.7, +10.7]	+4.2	[−4.6, +17.1]

pp: percentage-point difference in reliance. %: relative change from no dropout; zero denotes no change. Individual-seed contrasts appear in Table II.

TABLE IV

EXISTING CONSISTENCY-NOISE PAIR. EACH PAIR GIVES AMPLIFICATION FOR TRAINING EPISODES 0 / 60, IN THAT ORDER. ONE TRAINING SEED; NO POPULATION INTERVAL.

Limb	Without noise	Consistent noise
Left arm	3.79 / 2.63	0.93 / 1.01
Right arm	4.71 / 2.39	0.84 / 1.01
Left hand	2.46 / 2.42	1.08 / 1.03
Right hand	8.39 / 2.62	0.93 / 1.00

The action-parameterization effect depends on architecture. The one-seed ACT comparison shows substantial image response with absolute targets: at chunk length 100, ρ_{vis} is 2.04 and 2.35 across the two checkpoints. Relative targets fall below the historical fivefold visual-revival criterion. Replay comparisons favor absolute targets in 8/8 cells at chunk length 16 and 4/8 at length 100, across checkpoint and temporal-ensembling settings. These repeated measurements demonstrate that absolute-target policies can already be image-responsive and that the measured effect of target subtraction varies with architecture and configuration.

VI. DISCUSSION AND CONCLUSION

Why diagnose blindness? Success in familiar conditions leaves unresolved whether commands follow the scene or joint-state correlations. A displaced object may require a different action at the same posture. Blindness must be assessed against task-relevant changes and execution phase; invariance to irrelevant imagery can be desirable. Our scores diagnose input dependence; appropriate correction also requires evaluating response direction, timing, and consequence.

Implications for hierarchical control. Hi Robot separates high-level reasoning from low-level action [47]; GR00T N1 couples a vision-language module to a motor-action generator [48]. DuoCore-FS combines asynchronous semantic updates with fast visual/proprioceptive control [49]. Our findings suggest a criterion for this functional “big-brain/small-brain” division: preserve state feedback for local correction while checking that scene changes and updated subgoals influence commands. Accurate scene interpretation upstream

need not influence the action policy. Separate visual and subgoal interventions could locate this loss of influence. For bottle opening, cap separation should switch the subgoal from twisting to lifting while local feedback regulates motion. These are design implications; hierarchical policies remain untested here.

A human-action analogy. Human reaching can adjust to a displaced visual target without conscious awareness of its displacement [50]. Automatic execution can thus retain visual feedback. This motivates testing scene-dependent correction alongside familiar-sequence completion. The analogy concerns functional coordination, not cerebral or cerebellar anatomy. Our dropout results show why encouraging visual response must also account for command fidelity and feedback error.

Scope and conclusion. Historical diagnostic provenance is incomplete; donor substitutions can be off-distribution, windows emphasize transitions, and bootstrap intervals condition on trained models and donors. Noise and ACT comparisons use one training seed. Replay fixes external observations and measures numerical error without contact or servo dynamics. Diagnose input use first, then assess visual response, command fidelity, and feedback error jointly.

APPENDIX I

IMPLEMENTATION AND REPRODUCIBILITY

Training. All nine primary models use one observation, three 224×224 RGB views, a ViT-small R26/S32 encoder, a six-layer/four-head action transformer, and 16-step chunks. Training uses batch size 32, 200 epochs, AdamW (learning rate and weight decay 10^{-4} ; betas 0.95/0.999), cosine decay, 1,000 warmup steps, and crop ratio 0.95. The visual encoder is trained throughout; EMA and consistency noise are disabled. Seeds 42/43/44 share split seed 42. Checkpoints 99 and 198 are designated in advance.

For target i and 20 Gaussian candidates j , let d_{ij} be flattened normalized Euclidean distance and d_{max} its batch maximum. Training minimizes mean $b_i = \min_j \{d_{ij} + d_{\text{max}} \mathbf{1}[d_{ij} \leq 0.03]\}$ over $b_i < d_{\text{max}}$, or zero for an empty set. Relative subtraction precedes affine normalization fitted to training absolute commands. Inference uses one Gaussian latent per paired draw without candidate-selection history.

APPENDIX II

EVALUATION AND COMPLETE NUMERICAL RESULTS

Each held-out episode supplies three windows ranked by $\sum_j |A_{t+1,j} - A_{t,j}|$, separated by more than eight frames. This mixed-unit score can favor arm motion. Donors cycle within each held-out store, matching episode-progress deciles; camera views are swapped jointly. Inference seeds are 12345/999. Replay starts at frame five, selects command eight, advances eight frames, and resets each seed before every call for 25 replays.

The archived residual is $w(u) = ((u + 180) \bmod 360) - 180$, followed by limb RMS. Software uses arm degrees and hand radians, with incomplete collection-time unit/version binding; w is therefore a numerical, not physical-angle, score. An ordinary-residual check limited to six earlier seed-42/checkpoint-99 captures leaves hand scores and $G \geq 3$ fractions unchanged, but changes six arm episode-cell ratios.

Input-response and command-RMSE endpoints are averaged across training seeds after per-model reduction. Replay ratios are averaged over inference seeds, then median-reduced across episodes within a model, then averaged across training seeds. Peak-error summaries apply the same reductions to maxima. No evaluated denominator is zero. The bootstrap jointly resamples episode indices across models and training seeds (2,000 draws; seed 20260905). The frozen selection rule requires a reliance gain at both checkpoints, a final difference interval above zero, RMSE point ratios at most 1.10, no limb median- G ratio above 1.25, and no flag-rate increase above 10 percentage points. These operational criteria and all 36 cells' endpoints and paired intervals are independently reproduced from archived arrays.

Table V includes the untrained branches. Missing-state R_{vis} is omitted because the state donor is ignored; full-state zero/zero responses use the archived value zero. The no-dropout/missing-state error is a distribution-shift stress test.

Table VI summarizes episode maxima; dividing its columns does not reconstruct G , which aggregates episode-wise ratios. Command RMSE uses demonstrated commands; replay uses future recorded states, which may differ because of tracking lag.

To make the order of reductions explicit, index training seeds by s , episodes by i , and inference draws by r . For each limb, the reported endpoint is

$$\hat{G}_p = \frac{1}{3} \sum_s \text{median}_i \left[\frac{1}{2} \sum_{r=1}^2 G_{p,s,i,r} \right]. \quad (7)$$

Contrasts use unrounded aggregate endpoints, not per-seed ratios. Bootstrap replicates repeat the full reduction with shared episode indices and fixed donors. Intervals are point-wise across limbs.

Table VII gives final limbwise intervals; under M, only the right-hand interval excludes one.

APPENDIX III

EXPLORATORY AND COMPLEMENTARY PROTOCOLS

Exploratory diagnostics. Historical probes have incomplete checkpoint, split, and raw-array provenance. Arm gain

divides output displacement by imposed state displacement; the cross-limb effect uses the archived normalized read-out. Neither is a feedback eigenvalue. Tangent/orthogonal gains are 0.222/0.255, and a 2–16° amplitude sweep is approximately flat: these do not support strong anisotropy or saturation over that range. The cross-limb interval spanning zero does not establish independence.

The right glove has 24×32 taxels, one stuck channel (420), and 219 near-dead channels (28.5%). Contact windows use masked robust pressure aggregation. Contact-onset spread is across demonstrations, with onset-time regression $R^2 = 0.03$ (Table I). This excludes zero variation, not inadequate scene coverage. The left hand has no tactile sensor; its 0.61 enrichment measures event alignment, not tactile response or physical anticorrelation.

Recovered consistency-noise pair. The noise pair uses the exploratory collection (Section III) with head RGB and right-glove pressure, sharing training seed, augmentation, batch size, and normalizers. It uses checkpoint 100 and training episodes 0/60; the noise-sampler revision is not bound to the checkpoint. The probe averages target-chunk RMS changes over 120 training frames (image donor 105388, state offset 40, fixed pressure), then divides image by state response. Paired inference seeds are 12345/999; replay uses ordinary differences and seed 12345.

ACT comparison. ACT uses head RGB and state, one observation, chunk lengths 16/100, the LeRobot 0.4.4 CVAE, 100,000 steps, batch size eight, seed 12345, and a 179/20 split. Relative targets subtract the observation state from every step; target-specific normalizers differ. Evaluation uses latent mean zero, training-frame probes, midpoint/final checkpoints, and both ensembling settings. Its raw training losses are not a matched-fit comparison.

Historical cutoffs were $\rho_{\text{vis}} < 0.3$ for weak response and a fivefold relative/absolute increase for revival; they are not applied across protocols.

APPENDIX IV

ANALYTICAL DETAILS

Missing-state recurrence and finite-horizon gain. Using $Q_{k+1} = Q_k + u_k$, add and subtract Q_k^* to obtain

$$\begin{aligned} e_{k+1} &= Q_k + u_k - Q_{k+1}^* \\ &= e_k + (Q_k^* + u_k - Q_{k+1}^*) = e_k + \eta_k. \end{aligned} \quad (8)$$

Iteration with $e_0 = 0$ gives Eq. (6). The ordinary limb RMS norm and triangle inequality imply

$$\max_{1 \leq k \leq T} \|e_k\|_\ell \leq T \max_{0 \leq j < T} \|\eta_j\|_\ell. \quad (9)$$

Constant residuals give $e_k = k\eta$ and amplification T despite an identity Jacobian; alternating residuals may cancel. Large ratios need not imply multiplicative instability. The bound is per limb/draw in ordinary, unwrapped coordinates.

What target subtraction preserves. If $f_D = f_A - Q$, absolute and relative command maps, and their state derivatives, are identical. Under exact subtraction, $A \mapsto A - Q$ is bijective conditional on Q . At fixed $Q = q$, the inverse

TABLE V

COMPLETE DESIGN: 36 CELLS REDUCED TO 12 MEANS OF THREE TRAINING-SEED ENDPOINTS. F/M: FULL/MISSING STATE; *: UNTRAINED BRANCH. G IS THE MEAN OF THREE EPISODE MEDIANS. LA/RA/LH/RH: LEFT/RIGHT ARM/HAND. MISSING-STATE RELIANCE IS OMITTED.

Epoch	p	Mask	R_{vis}	RMSE	G_{LA}	G_{RA}	G_{LH}	G_{RH}
99	0	F	0.417	0.05838	2.45	2.20	2.34	1.89
99	0	M*	–	0.13022	5.15	5.02	16.52	3.76
99	0.5	F	0.988	0.06354	3.99	2.36	15.71	1.80
99	0.5	M	–	0.06374	4.02	2.34	16.01	1.80
99	1	F*	0.999	0.06367	4.18	2.54	16.18	1.71
99	1	M	–	0.06367	4.18	2.54	16.18	1.70
198	0	F	0.392	0.05654	2.25	2.19	2.09	1.68
198	0	M*	–	0.14456	4.71	5.30	16.05	3.64
198	0.5	F	0.990	0.06127	4.04	2.34	15.97	1.76
198	0.5	M	–	0.06147	4.03	2.34	16.12	1.75
198	1	F*	0.999	0.06106	3.95	2.31	16.04	1.64
198	1	M	–	0.06106	3.96	2.31	16.04	1.64

TABLE VI

EPOCH-198 ERROR MAGNITUDES UNDER THE FROZEN REPLAY METRIC. PER EPISODE, EACH COLUMN AVERAGES TWO INFERENCE-SEED MAXIMA; THE TABLE THEN REPORTS THE MEAN OF THREE TRAINING-SEED EPISODE MEDIANS. TEACHER/AUTO DENOTE RECORDED-STATE/AUTOREGRESSIVE INPUT.

p / Mask	Limb	Teacher	Auto
0 / F	LA	8.672	18.617
	RA	8.578	17.317
	LH	0.300	0.557
	RH	0.131	0.204
0.5 / F	LA	8.786	34.679
	RA	8.729	18.489
	LH	0.305	5.280
	RH	0.137	0.224
0.5 / M	LA	8.811	34.301
	RA	8.728	18.288
	LH	0.306	5.355
	RH	0.138	0.224
1 / M	LA	8.642	35.544
	RA	8.464	17.171
	LH	0.303	5.332
	RH	0.143	0.221

is $D \mapsto D + q$. Mutual information is invariant under this invertible translation; rounding, clipping, or requantization need not preserve the identity. Hence

$$I(D; V \mid Q) = I(A; V \mid Q). \quad (10)$$

Withholding encoder state changes conditioning; differences between trained maps involve learning or approximation beyond this identity.

Finally, take orthogonal whitened features q, z , target $a = q + \Delta^*(z)$, $\mathbb{E}[\Delta^* q^\top] = 0$, and $\mathbb{E}[\Delta^* z^\top] = \epsilon B$. Linear absolute/relative predictors with population squared loss and zero initialization have state gradients $-2I/0$, but identical visual gradients. Gradient descent gives identical visual weights $W_z^{(k)} = U_z^{(k)} = [1 - (1 - 2\eta)^k] \epsilon B$. This linear counterexample separates removal of the initial state-gradient block from acceleration of visual learning. Its squared-loss

TABLE VII

FINAL-CHECKPOINT AMPLIFICATION CONTRASTS WITH CONDITIONAL 95% PAIRED EPISODE-BOOTSTRAP INTERVALS. RATIOS COMPARE MEANS OF TRAINING-SEED ENDPOINTS.

Comparison	Limb	Ratio [95% interval]
0.5 vs. 0 (F)	LA	1.797 [1.645, 1.977]
	RA	1.070 [0.963, 1.167]
	LH	7.636 [4.972, 8.286]
	RH	1.042 [0.954, 1.171]
1 vs. 0.5 (M)	LA	0.983 [0.946, 1.041]
	RA	0.989 [0.920, 1.106]
	LH	0.995 [0.974, 1.012]
	RH	0.939 [0.898, 0.988]

assumptions differ from the generative training objective used in the experiment.

Consistency under state noise. For the recomputed target, the samplewise command residual is

$$(\tilde{Q} + \hat{D}_h) - A_h = \hat{D}_h - \tilde{D}_h. \quad (11)$$

With a common squared metric, recomputed-target and command errors coincide. Retaining the old target shifts an exact-target command by ξ . Zero-mean noise still changes the samplewise objective. The checkpoint pair does not isolate target recomputation.

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